

Changbeom Shim, Ph.D.

4.14, New Technologies Bld. (314) ◇ Kent St, Bentley ◇ Western Australia, 6102, Australia

Research Fellow ◇ School of Elec. Eng., Comp. and Math. Sci. ◇ Curtin University

Education

Korea University

Ph.D. in Computer Science and Engineering

· Supervisor: Prof. Yon Dohn Chung

Seoul, South Korea

Mar. 2014 - Aug. 2019

Korea University

B.E. in Electronics and Information Engineering

· On leave: SEP. 2008 - FEB. 2011, Military Service

Sejong, South Korea

Mar. 2007 - Feb. 2014

Professional Experiences

School of Elec. Eng., Comp. & Math. Sci. at Curtin University

Research Fellow

Research Associate

Perth, Australia

Nov. 2022 - Present

Nov. 2019 - Nov. 2022

Database Lab. at Korea University

Research Assistant

Seoul, South Korea

Mar. 2014 - Aug. 2019

School of Engineering at RMIT University

Visiting Student (*Host*: Dr. Du Yong Kim)

Melbourne, Australia

Jan. 2019 - Feb. 2019

Funded Projects

South Korea: Yon Dohn Chung

2024 - 2025

Australia: **Changbeom Shim**

“Privacy-preserving processing and sharing of spatio-temporal data”

₩20,000K

International Mobility Program

National Research Foundation of Korea, South Korea

Australia: **Changbeom Shim**, Ba-Ngu Vo, Ba Tuong Vo, Hoa Van Nguyen, Tran Thien Dat Nguyen

2024 - 2025

South Korea: Yon Dohn Chung, Hyunjin Choi, Yeonghun Chae

“Enhancing Australia-Korea emergency management capabilities”

A\$23K

Australia-Korea Foundation Grant

Australia-Korea Foundation, Australia

Changbeom Shim, Ba Tuong Vo, Youngho Kim, Jin Bum Hong, Sanjeev Arulampalam, Du Yong Kim

2023 - 2025

“Scalable sensor fusion for 3D environmental mapping”

A\$150K

Collaborative Research Grant

Defence Science Centre, Australia

Changbeom Shim

2020 - 2021

“An intelligent system for crime analysis and prevention based on multi-object tracking and situational awareness” ₩45,000K

Postdoctoral Fellowship (Nurturing Next-generation Researchers) National Research Foundation of Korea, South Korea

Honors and Awards

NOV. 2022 Best Paper Award, Int. Conf. Cont., Auto. and Inf. Sci. (ICCAIS)

OCT. 2021 Best Paper Award (First Runner-Up), Int. Conf. Cont., Auto. and Inf. Sci. (ICCAIS)

FEB. 2020 Global Talent Independent Program (Invitation to Data Science & ICT), The Australian Government

FEB. 2019 KU Graduate Student Achievement Award, Korea University

JUL. 2018 Student Paper Contest Award, Korea Computer Congress (KCC)

JUL. 2018 LG (Chief Technology Officer) Scholarship, LG Electronics

FEB. 2018 Outstanding Ph.D. Candidate Paper Award, Korea University

OCT. 2017 Honorable Mention Award, IEEE Visual Analytics Science and Technology (VAST Challenge)

MAY 2015 Outstanding Paper Award, Korean DataBase Conference (KDBC)

International Journals

- J15** Tran Thien Dat Nguyen, Ba Tuong Vo, Ba-Ngu Vo, Hoa Van Nguyen, **Changbeom Shim**
“The Mean of Multi-Object Trajectories”
IEEE Transactions on Signal Processing, In Press, Feb. 2026
- J14** **Changbeom Shim**, Jangwon Gim, Yeeun Kim, Yeonghun Chae
“DIZEST: A low-code platform for workflow-driven artificial intelligence and data analysis”
SoftwareX, Vol 33: 102519, Jan. 2026
- J13** Linh Van Ma, Tran Thien Dat Nguyen, **Changbeom Shim**, Du Yong Kim, Namkoo Ha, Moongu Jeon
“Visual multi-object tracking with re-identification and occlusion handling using labeled random finite sets”
Pattern Recognition, Vol 156: 110785, Dec. 2024
- J12** Ba-Ngu Vo, Ba-Tuong Vo, Tran Thien Dat Nguyen, **Changbeom Shim**
“An Overview of Multi-Object Estimation via Labeled Random Finite Set”
IEEE Transactions on Signal Processing, Vol 72: 4888-4917, Oct. 2024
- J11** **Changbeom Shim**, Ba-Tuong Vo, Ba-Ngu Vo, Jonah Ong, Diluka Moratuwage
“Linear Complexity Gibbs Sampling for Generalized Labeled Multi-Bernoulli Filtering”
IEEE Transactions on Signal Processing, Vol 71: 1981-1994, May 2023 (also presented in ICASSP 2024)
- J10** **Changbeom Shim**, Wooil Kim, Tran Thien Dat Nguyen, Du Yong Kim, Yu Suk Choi, Yon Dohn Chung
“CellTrackVis: interactive browser-based visualization for analyzing cell trajectories and lineages”
BMC Bioinformatics, Vol 24: 124, Mar. 2023
- J9** Diluka Moratuwage, Ba-Ngu Vo, Ba-Tuong Vo, **Changbeom Shim**
“Multi-Sensor Multi-Scan Multi-Object State Estimation”
IEEE Transactions on Signal Processing, Vol 70: 5429-5442, Oct. 2022
- J8** Cong-Thanh Do, Tran Thien Dat Nguyen, Diluka Moratuwage, **Changbeom Shim**, Yon Dohn Chung
“Multi-object tracking with an adaptive generalized labeled multi-Bernoulli filter”
Signal Processing, Vol 196: 108532, Jul. 2022
- J7** Jonah Ong, Ba Tuong Vo, Sven Nordholm, Ba-Ngu Vo, Diluka Moratuwage, **Changbeom Shim**
“Audio-Visual Based Online Multi-Source Separation”
IEEE/ACM Transactions on Audio, Speech, Language Processing, Vol 30: 1219-1234, Mar. 2022
- J6** Wooil Kim, **Changbeom Shim**, Yon Dohn Chung
“SkyFlow: A visual analysis of high-dimensional skylines in time-series”
Journal of Visualization, Vol 24(5): 1033-1050, May 2021
- J5** **Changbeom Shim**, Gyuhyeon Sim, Yon Dohn Chung
“Cohesive Ridesharing Group Queries in Geo-Social Networks”
IEEE Access, Vol 8: 97418-97436, May 2020
- J4** Wooil Kim, **Changbeom Shim**, Wan Heo, Sungmin Yi, Yon Dohn Chung
“Moving view field nearest neighbor queries”
Data & Knowledge Engineering, Vol 119: 58-70, Jan. 2019
- J3** **Changbeom Shim**, Wooil Kim, Wan Heo, Sungmin Yi, Yon Dohn Chung
“Nearest close friend search in geo-social networks”
Information Sciences, Vol 423: 235-256, Jan. 2018
- J2** Sungmin Yi, **Changbeom Shim**, Yon Dohn Chung
“Reverse View Field Nearest Neighbor Queries”
Information Sciences, Vol 402: 35-49, Sep. 2017
- J1** **Changbeom Shim**, Wooil Kim, Wan Heo, Sungmin Yi, Yon Dohn Chung
“ ℓ -Close Range Friends Query on Social Grid Index”
IEICE Transactions on Information and Systems, Vol 100-D(4): 811-812, Apr. 2017

International Conferences

- C14** Tran Thien Dat Nguyen, Hoa Van Nguyen, **Changbeom Shim**
“Multi-Sensor Multi-Object Trajectory Estimation with Lineage Inference”
14th International Conference on Control, Automation and Information Sciences (ICCAIS), Oct. 2025
- C13** Chaerin Kim, Yeonghun Chae, Hyoungki Ahn, **Changbeom Shim**
“Korean Speech Recognition via Retrieval-Augmented Generation-Based Intent-Aware Keyword Correction”
14th International Conference on Control, Automation and Information Sciences (ICCAIS), Oct. 2025
- C12** Hoa Van Nguyen, Diluka Moratuwage, Tran Thien Dat Nguyen, **Changbeom Shim**, Ba-Ngu Vo
“Multi-Agent Planning for Pursuing Multiple Objects via Proximal Policy Optimisation”
14th International Conference on Control, Automation and Information Sciences (ICCAIS), Oct. 2025
- C11** Bohyun Hwang, **Changbeom Shim**, Seonggun Joe, Byungkyu Kim
“A Preliminary Study on Autonomous Robotic Colonoscopy via Deep Neural Network”
13th International Conference on Control, Automation and Information Sciences (ICCAIS), Nov. 2024
- C10** Tran Thien Dat Nguyen, Hoa Van Nguyen, **Changbeom Shim**, Viet Nguyen, Hyunjin Choi
“Efficient 3D Multi-View Multi-Object Tracking with Labeled Multi-Bernoulli Approximation”
13th International Conference on Control, Automation and Information Sciences (ICCAIS), Nov. 2024
- C9** **Changbeom Shim**, Youngho Kim, Craig Butterworth
“Property Estimation in Geotechnical Databases Using Labeled Random Finite Sets”
13th International Conference on Control, Automation and Information Sciences (ICCAIS), Nov. 2024
- C8** Hoa Van Nguyen, Tran Thien Dat Nguyen, **Changbeom Shim**, Diluka Moratuwage, Seonggun Joe
“Label Management for Multi-Object Estimates via Labelled Multi-Bernoulli Filters”
13th International Conference on Control, Automation and Information Sciences (ICCAIS), Nov. 2024
- C7** Ji Youn Lee, **Changbeom Shim**, Hoa Van Nguyen, Tran Thien Dat Nguyen, Hyunjin Choi, Youngho Kim
“Label Space Partition Selection for Multi-Object Tracking Using Two-Layer Partitioning”
12th International Conference on Control, Automation and Information Sciences (ICCAIS), Nov. 2023
- C6** Hoa Van Nguyen, Tran Thien Dat Nguyen, **Changbeom Shim**, Marzhar Anuar
“The Smooth Trajectory Estimator for LMB Filters”
12th International Conference on Control, Automation and Information Sciences (ICCAIS), Nov. 2023
- C5** **Changbeom Shim**, Ji Youn Lee, Diluka Moratuwage, Du Yong Kim, Yon Dohn Chung
“Generalized Label Grouping for Scalable Trajectory Estimation”
11th International Conference on Control, Automation and Information Sciences (ICCAIS), Nov. 2022
- C4** Diluka Moratuwage, **Changbeom Shim**, Yuthika Punchihewa
“A Moving Window Based Approach to Multi-Scan Multi-Target Tracking”, **Best Paper Award**
11th International Conference on Control, Automation and Information Sciences (ICCAIS), Nov. 2022
- C3** **Changbeom Shim**, Du Yong Kim
“Space-oriented Label Partitioning for Multi-object Tracking”, **Best Paper Award (First Runner-Up)**
10th International Conference on Control, Automation and Information Sciences (ICCAIS), Oct. 2021
- C2** Tran Thien Dat Nguyen, **Changbeom Shim**, Wooil Kim
“Biological Cell Tracking and Lineage Inference Via Random Finite Sets”
18th IEEE International Symposium on Biomedical Imaging (ISBI), Apr. 2021
- C1** Wooil Kim, **Changbeom Shim**, Ilhyun Suh, Yon Dohn Chung
“A Visual Explorer for Analyzing Trajectory Patterns”, **Honorable Mention Award (Challenge)**
12th IEEE Conference on Visual Analytics Science and Technology (VAST), Oct. 2017

Patents

- P2** Yon Dohn Chung, **Changbeom Shim**, Wooil Kim
“Device for searching nearest close friends”
Oct. 2020, Korea Registration Number: 10-2172236
- P1** Yon Dohn Chung, **Changbeom Shim**
“Device and method for recommending ride sharing group, recording medium for performing the method”
Jan. 2020, Korea Registration Number: 10-2064376

Supervisions

Namil Kim	Ph.D. student from Oct. 2024 under my co-supervision with Prof. Yon Dohn Chung
Yujeong Jo	Ph.D. student from Oct. 2024 under my co-supervision with Prof. Yon Dohn Chung
Ji Youn Lee	Ph.D. student from Jun. 2022 under my co-supervision with Prof. Ba Tuong Vo and Prof. Ba-Ngu Vo

Teaching

At Curtin University, Australia

S1 2026	CMPE3006: Advanced Digital Design	UNIT CO-ORDINATOR & LECTURER
S2 2025	EEET4001: Engineering Research Project 2	PROJECT SUPERVISOR
S1 2025	EEET4000: Engineering Research Project 1	PROJECT SUPERVISOR

At Korea University, South Korea

S2 2018	COSE214: Algorithm	LECTURER
S2 2017	COSE156: Python Programming	TEACHING ASSISTANT
S1 2016	COSE213: Data Structure	TEACHING ASSISTANT
S2 2015	COSE242: Data Communication	TEACHING ASSISTANT
S2 2014	COSE102: Computer Programming	TEACHING ASSISTANT

Professional Services

Committees	ICCAIS 2025 (Publication Chair), ICCAIS 2024 (Program Chair), ICCAIS 2023 (Special Session Chair)
Reviewers	IEEE Trans. Signal Process., IEEE Trans. Multim., IEEE Trans. Aerosp. Electron. Syst., Inf. Fusion, Neurocomputing, Digit. Signal Process., IEEE Access, J. Ambient Intell. Humaniz. Comput., Comput. Elec. Agri., Sensors, J. Imaging, ICCAIS (2022-2025)

Talks

OCT. 2025	INVITED TALK	Sejong University, Seoul, South Korea
FEB. 2025	SEMINAR TALK	Korea University, Seoul, South Korea
FEB. 2025	SEMINAR TALK	Korea Aerospace University, Goyang, South Korea
JUN. 2023	INVITED TALK	Sookmyung Women's University, Seoul, South Korea
DEC. 2020	SEMINAR TALK	Korea University, Seoul, South Korea

References

References are available upon request.